# (10) Multi-modal Rags

## (V1) → Retrieving and Integrating Information in Rag system

→ A multimodal Rag model is an advanced type of neural network that processes and generate information from multiple types of data modalities such as text, images, and videos.

→ Prepare Text, image, and video Data

→ We can design a model to help student with their ~~student queries~~ system can leverage text, image and video data to provide comprehensive and contextually rich answers to students queries.

## (V2) → Apply PCA to Text, image and Video Embedelings

→ PCA is a powerful tool for dimensionlity reduction it helps to simple complex data sets by focusing on the most important aspects of the data we need to get text, image and video embeddings.

## (V3) → Combining Text, Image and Video Embeddings

→ The purpose of combining Embeddings from different modalities is to create a unified representation of the data that captures information from all sources. By stacking the embedding vertically, we concatenate them row wise, allowing us to maintain the individual embeddings for each modality while creating a single array that contains all the information.

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## (V4) Create Fais index and Query the index.

→ Fais stands for Facebook AI similarity search

→ It is an open source library developed by Facebook. In rag system, fais plays a crucial role in the efficient and scalable retrieval of relevant documents, images and videos

→ The purpose of creating fais index is to perform a similarity search.

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